

Are You Getting What You Pay For? Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

"Commercial LLM API providers may covertly substitute cheaper models"

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

H_0 : Specified Model

vs

H_1 : Substituted Model

Can auditor distinguish $P_{\text{spec}}(y|x)$ from $P_{\text{actual}}(y|x)$?



Smaller Model Substitution

Different model size tier

Example: 8B served instead of 70B

Cost reduction: ~85%



FP8/INT8 Quantization

Lower precision inference

Example: FP16 $\rightarrow$ FP8 weights

Cost reduction: ~40-60%



Fine-tuned Variant Swap

Different training iteration

Example: Updated checkpoints

Harder to detect without benchmarks



Different Model Family

Entirely different architecture

Example: Qwen served as Llama

Significant behavioral differences

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond	Status
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22	Original
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24	FP8
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51	Original
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30	FP8
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97	Original
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22	FP8

Key Insight: Scores fall within error bars — detection is extremely difficult without large sample sizes

Higher Temperature Amplifies Performance Gaps but Increases Noise

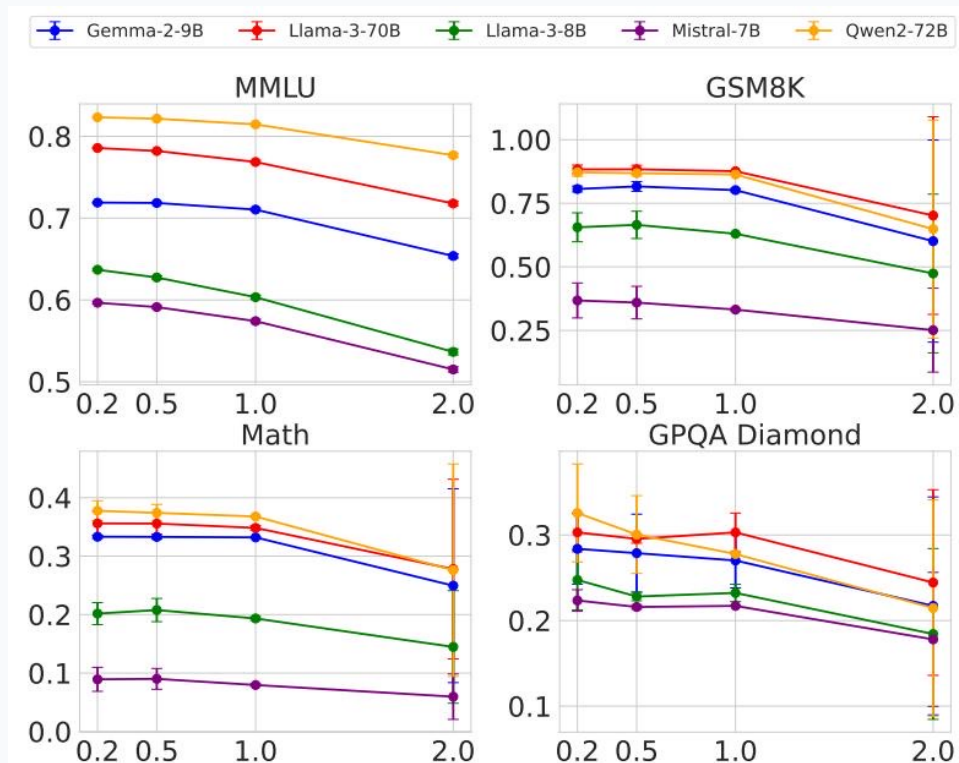


Figure: Benchmark accuracy vs temperature for five models

Temperature Scaling Effect

- At temperature 0.2: Original and FP8 models show clear performance separation
- At temperature 1.0: Gap narrows but variance increases
- At temperature 2.0: Performance degrades across all models
- Gap between original and substitute becomes noisy and inconsistent

Detection Challenge:

Higher temperatures obscure substitution signals. Statistical tests require impractically large samples.

Four benchmarks tested: MMLU, GSM8K, MATH, GPQA Diamond

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure: Log probability variation across inference frameworks and hardware for Llama-3-8B

Production-Level Nondeterminism

The same Llama-3-8B model produces different log probabilities based on:

Software Framework:

- transformers 4.40 vs 4.50
- vLLM 0.6.2 vs 0.8.2

GPU Hardware:

- NVIDIA A100 vs H100

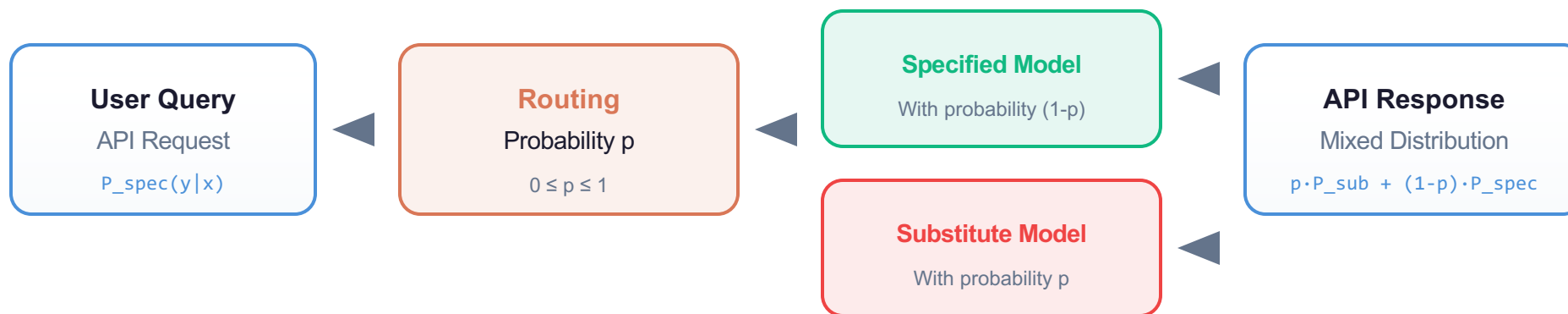
Why This Defeats Detection:

An auditor cannot distinguish between:

- ① A legitimate model served on different infrastructure
- ② An actual model substitution

→ **Log-probability fingerprinting is unreliable in practice**

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive



Key Mathematical Insight:

As $p \rightarrow 0$: $p \cdot P_{\text{sub}} + (1-p) \cdot P_{\text{spec}} \approx P_{\text{spec}}$

The mixed distribution converges to the specified model's distribution, requiring exponentially more queries for detection.

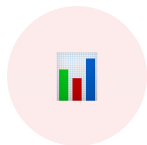
Query Complexity

- At $p = 0.5$: ~1,000 queries needed for detection
- At $p = 0.1$: ~100,000 queries needed
- **At $p = 0.01$: Exponentially more — practically infeasible**

Adversarial Advantage

- Provider controls the probability p
- Can dynamically adjust to evade detection
- "Query-prohibitive" = economically infeasible to audit

Software-Only Auditing Methods Are Fundamentally Unreliable



Text-based Statistical Tests

e.g., Maximum Mean Discrepancy (MMD)

METHOD TYPE

Distribution comparison via text outputs

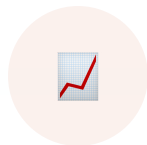
WEAKNESS

Requires impractically large query budgets; fails against

PRACTICAL LIMITATION

Detection requires 100K+ queries at low substitution rate

FAILS



Log-Probability Methods

Token-level fingerprinting

METHOD TYPE

Compare log probabilities between local reference and

WEAKNESS

Production inference nondeterminism across software versions

PRACTICAL LIMITATION

Cannot distinguish legitimate infrastructure differences

DEFEATED



Adversarial Countermeasures

Active evasion strategies

METHOD TYPE

Provider-controlled routing: randomized substitution + benchmark

WEAKNESS

Makes all software-only detection fundamentally query-prohibitive

PRACTICAL LIMITATION

Auditor cannot detect known benchmark queries routed to geographically

UNAVOIDABLE

Conclusion: All software-based approaches share a common flaw — they observe outputs, not execution.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity



Provable

Cryptographic attestation guarantees
correct model weights are loaded

Hardware-signed proofs



Efficient

Modest performance overhead —
practical for production deployment

vs. ZKPs (prohibitively expensive)



Robust

No dependency on observing outputs
or inference behavior

Resistant to adversarial evasion

How TEE Attestation Works

1. Hardware creates isolated enclave environment
2. Model weights loaded and measured by hardware
3. Attestation quote provides cryptographic proof
4. Auditor verifies quote matches expected model

Practical Deployment Path

NVIDIA Confidential Computing on H100 GPUs

- Hardware-level security built into GPU
- Production-ready today

✓ Available for immediate adoption

KEY TAKEAWAYS

Close the Verification Gap with Hardware Attestation

1

Model substitution is economically motivated and practically undetectable via software

FP8 quantization barely affects benchmarks; cost incentives drive covert swaps

2

FP8 quantization preserves benchmark performance within error bars

Detection requires impractically large sample sizes and query budgets

3

Inference nondeterminism kills log-probability fingerprinting in production

Software version and GPU differences produce varying log probabilities

4

Randomized substitution defeats all statistical detection approaches

As $p \rightarrow 0$, mixed distribution converges to specified model — query-prohibitive

5

TEEs are the only viable path to verifiable LLM APIs today

Hardware attestation provides provable, efficient, robust model integrity

Call to Action: Advocate for hardware attestation adoption in commercial LLM APIs